# Quantitative Reasoning Worksheet

*Lab 6: Quantitative Reasoning I*

**Team members:** _____

**Case number and title:** _____

*Instructions: Use your team's case scenario. Carry units in [square brackets] and state the time basis for every number. Show formulas and working, not only answers.*

## 1. Process Flow (4–6 stages)

*Write the stages in order, left to right, then sketch arrows between them.*

## 2. Demand and Capacity

*Same unit and same time basis on both sides (e.g. units/day).*

| Stage | Capacity (unit / h) | Time basis (h/day) | Capacity (unit / day) |
|---|---|---|---|
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Average demand: _____ [unit / _____]  Peak demand: _____ [unit / _____]

## 3–4. Utilisation and Classification

*Utilisation = Demand ÷ Capacity.  Comfortable < ≈0.8 · Stressed ≈0.8–1.0 · Over capacity > 1.0*

| Stage | Util. at average | Util. at peak | Comfortable / Stressed / Over capacity (at peak) |
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## 5. Bottleneck and Maximum Practical Throughput

Bottleneck stage: _____ Maximum throughput: _____ [unit / _____]

*Why is this the bottleneck?*

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## 6. Material Flow: Inflow, Outflow, Rejects, Losses, Rework, Scrap

| After stage | Loss / reject rate | Units in | Units lost | Units out |
|---|---|---|---|---|
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Good final output per day: _____ [unit/day] Share lost: _____ %

## 7. Queue, Buffer or Stock Change

*Change = inflow − outflow.*

| Item | Your value (with unit) |
|---|---|
| Which queue / buffer / stock? | |
| Inflow [ / h ] | |
| Outflow [ / h ] | |
| Change = in − out [ / h ] | |
| Current level and maximum size | |
| Size after stated time, OR time to become full (show working) | |

## 8. Average vs Peak Demand

| | Demand | Bottleneck utilisation | Does the system cope? |
|---|---|---|---|
| Average | | | |
| Peak | | | |

*Does the system perform differently at peak? Explain.*

## 9. One Quantified Improvement

☐Reduce / shift demand ☐Increase capacity ☐Accept / manage waiting

| Item | Your answer |
|---|---|
| What exactly do you change? | |
| Target utilization | |
| Calculation (show working) | |
| New capacity or new utilization | |
| Side effect: which stage is now tightest? | |

## 10. Order-of-Magnitude / Reasonableness Check

Answer being checked: _____

| Check | Your result |
|---|---|
| Expected scale (tens / hundreds / thousands …) and quick estimate | |
| Units check: do the units and time basis work out? | |
| Is the result plausible? Why? | |